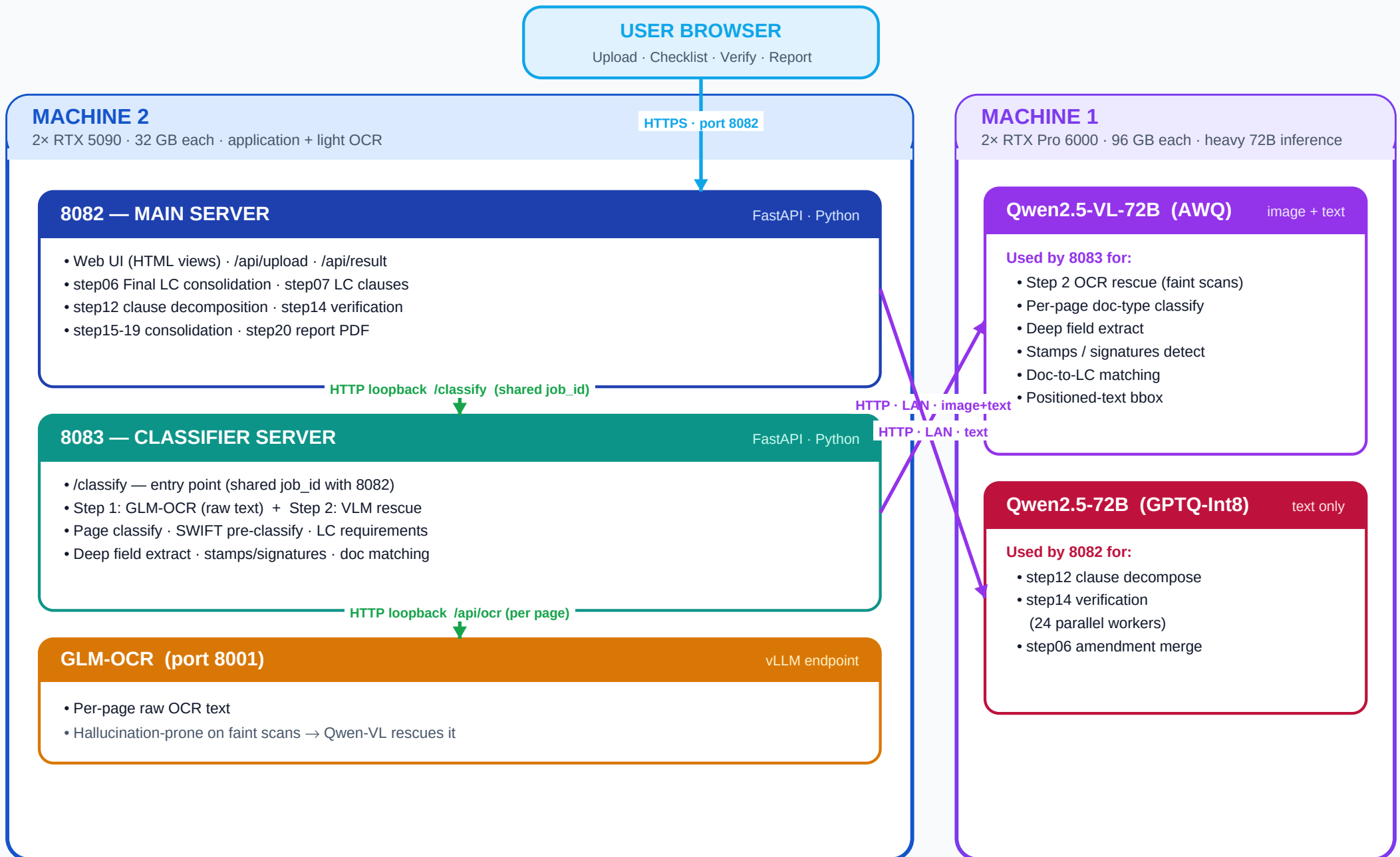


ComplyTrade — System Architecture

Two-machine deployment · loopback + cross-machine HTTP



Machine	Service	Owns
Machine 2	8082 main server	Upload UI · job state · step06 Final LC · step07 LC clauses · step12-14 verification fan-out · step15-20 report
Machine 2	8083 classifier	Per-page OCR (GLM + VLM rescue) · doc-type classify · SWIFT pre-classify · LC requirements parse · deep field extract · stamps · positioned-text
Machine 2	GLM-OCR (vLLM :8001)	Step 1 — raw per-page text. Lightweight, fits on 5090s.
Machine 1	Qwen2.5-VL-72B (AWQ)	Image+text reasoning. Step 2 OCR rescue · classification · deep_extract · matching · bbox.
Machine 1	Qwen2.5-72B (GPTQ-Int8)	Text-only LLM. Clause decomposition (step12) · verification (step14, 24 parallel) · amendment merge (step06).

Network endpoints

Machine 2 (apps)

8082 main server	http://M2:8082
8083 classifier	http://localhost:8083 (8082 → 8083 loopback)
GLM-OCR	http://10.20.10.2:8001/api/ocr (VPN local)
GLM-OCR (fallback)	http://34.171.200.116/api/ocr (GCP)

Machine 1 (heavy LLMs)

Qwen-VL 72B	http://M1:.../vllm/v1/chat/completions
Qwen-LLM 72B	http://M1:.../v1/chat/completions

Per-page OCR log tags (8082 live processing panel)

(no tag)	GLM was clean — used as-is
[text reader lines stripped]	GLM had prompt-template lines mixed in; stripped, rest kept
[text reader rescue ✓]	GLM hallucinated → Qwen-VL re-OCR'd from image; recovered text
[text reader blank]	GLM produced minimal output; page treated as blank
[text reader+trade expert blank]	Both GLM and Qwen-VL agree the page is blank

Phase 1 — Upload & Classification

- 1 **User uploads PDF**
Browser → 8082 (HTTPS, POST /api/upload)
- 2 **8082 generates job_id**
stores source.pdf in results/<job_id>/
- 3 **8082 calls classifier**
8082 → 8083 (HTTP loopback, /classify, same job_id)
- 4 **8083 runs OCR in parallel**
8083 → GLM-OCR (HTTP loopback, /api/ocr per page)
- 5 **Rescue when GLM fails**
8083 → Qwen-VL 72B (LAN, image + transcribe prompt)
- 6 **8083 classifies each page**
8083 → Qwen-VL 72B (LAN, image + GLM text)
- 7 **SWIFT + LC requirements**
8083 parses :20/:31C/:46A/:47A: regex on cleaned text
- 8 **Deep field extract**
8083 → Qwen-VL 72B (LAN, per packet, fields + stamps)
- 9 **Match shipping docs ↔ LC**
8083 → Qwen-VL 72B (LAN, doc page vs. LC requirement)
- 10 **Classifier returns to main**
8083 → 8082 (classification.json: pages, logical_docs, fields)
- 11 **Adapter → step01..step09**
8082 in-process: bridge/adapter.py converts to legacy shape
- 12 **Final LC consolidation**
8082 step06 (regex on OCR text → consolidated_fields, clauses)
- 13 **Clause + requirement extract**
8082 step07 (LC requirements + conditions parsed from F46A/F47A)

Phase 2 — Verification (on user request)

- 14 **User clicks Verify**
Browser → 8082 (POST /api/verify/{job_id}/{lc_number})
- 15 **Decompose clauses**
8082 step12 → Qwen-LLM 72B (LAN, ~24 parallel workers)
- 16 **Build verification rows**
8082 step13 in-process (rows = clause × target doc)
- 17 **Verify each row**
8082 step14 → Qwen-LLM 72B (LAN, parallel, doc text + audit header)
- 18 **Confidence + cross-clause**
8082 step15-17 in-process (overrides, escalations, dependencies)
- 19 **Consolidate**
8082 step18-19 in-process (final result rows)
- 20 **Report PDF**
8082 step20 generates compliance_report.pdf
- 21 **Browser displays result**
8082 → Browser (JSON results + report download URL)